

Instance-based Learning Classifiers/K-Nearest Neighbor Algorithm

This tutorial is an introduction to an instance based learning called K-Nearest Neighbor or KNN algorithm. KNN is part of supervised learning that has been used in many applications in the field of data mining, statistical pattern recognition, image processing and many others. Some successful applications are including recognition of handwriting, satellite image and EKG pattern. Instead of using sophisticated software or any programming language,

What is K-Nearest Neighbor (KNN) Algorithm?

Let us start with K-nearest neighbor algorithm for classification. K-nearest neighbor is a supervised learning algorithm where the result of new instance query is classified based on majority of K-nearest neighbor category. The purpose of this algorithm is to classify a new object based on attributes and training samples. The classifiers do not use any model to fit and only based on memory. Given a query point, we find K number of objects or (training points) closest to the query point. The classification is using majority vote among the classification of the K objects. Any ties can be broken at random. KNN used neighborhood classification as the prediction value of the new query instance.

Example

For example

We have data from the questionnaires survey (to ask people opinion) and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here is four training samples

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with $X1 = 3$ and $X2 = 7$. Without another expensive survey, can we guess what the classification of this new tissue is? Fortunately, KNN algorithm can help you to predict this type of problem.

How does KNN Work?

K nearest neighbor algorithm is very simple. It works based on minimum distance from the query instance to the training samples to determine the K-nearest neighbors. After we gather K nearest neighbors, we take simple majority of these K-nearest neighbors to be the prediction of the query instance.

The data for KNN algorithm consist of several multivariate attributes name X_i that will be used to classify Y . The data of KNN can be any measurement scale from ordinal, nominal, to quantitative scale but for the moment let us deal with only quantitative X_i and binary (nominal) Y . Later in this section, I will explain how to deal with other types of measurement scale.

Numerical Example

Example

We have data from the questionnaires survey (to ask people opinion) and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here is four training samples

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Y = Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with $X1 = 3$ and $X2 = 7$. Without another expensive survey, can we guess what the classification of this new tissue is?

1. Determine parameter K = number of nearest neighbors
2. Calculate the distance between the query-instance and all the training samples
3. Sort the distance and determine nearest neighbors based on the K -th minimum distance
4. Gather the category Y of the nearest neighbors
5. Use simple majority of the category of nearest neighbors as the prediction value of the query instance

Example Continued

1. Determine parameter K = number of nearest neighbors

Suppose use $K = 3$

2. Calculate the distance between the query-instance and all the training samples

Coordinate of query instance is (3, 7), instead of calculating the distance we compute square distance which is faster to calculate (without square root)

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)
7	7	$(7-3)^2 + (7-7)^2 = 16$
7	4	$(7-3)^2 + (4-7)^2 = 25$
3	4	$(3-3)^2 + (4-7)^2 = 9$
1	4	$(1-3)^2 + (4-7)^2 = 13$

3. Sort the distance and determine nearest neighbors based on the K -th minimum distance

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?
7	7	$(7-3)^2 + (7-7)^2 = 16$	3	Yes
7	4	$(7-3)^2 + (4-7)^2 = 25$	4	No
3	4	$(3-3)^2 + (4-7)^2 = 9$	1	Yes
1	4	$(1-3)^2 + (4-7)^2 = 13$	2	Yes

Example Continued

4. *Gather the category Y of the nearest neighbors.* Notice in the second row last column that the category of nearest neighbor (Y) is not included because the rank of this data is more than 3.

X1 = Acid Durability (seconds)	X2 = Strength (kg/square meter)	Square Distance to query instance (3, 7)	Rank minimum distance	Is it included in 3-Nearest neighbors?	Category of nearest Neighbor
7	7	$(7-3)^2 + (7-7)^2 = 16$	3	Yes	Bad
7	4	$(7-3)^2 + (4-7)^2 = 25$	4	No	-
3	4	$(3-3)^2 + (4-7)^2 = 9$	1	Yes	Good
1	4	$(1-3)^2 + (4-7)^2 = 13$	2	Yes	Good

5. *Use simple majority of the category of nearest neighbors as the prediction value of the query instance*

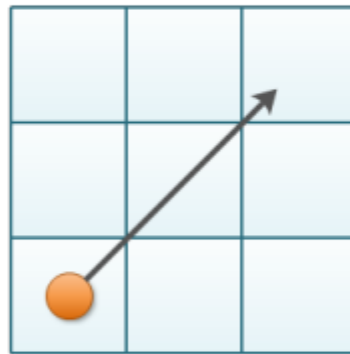
We have 2 good and 1 bad, since $2 > 1$ then we conclude that a new paper tissue that pass laboratory test with $X1 = 3$ and $X2 = 7$ is included in **Good** category.

Other Distance Metrics (Euclidean, Chebyshev, and Manhattan Distances)

There are many metrics to calculate a distance between 2 points $p(x_1, y_1)$ and $q(x_2, y_2)$ in xy -plane. We can count Euclidean distance, or Chebyshev distance or manhattan distance, etc. Each one is different from the others.

To simplify the idea and to illustrate these 3 metrics, I have drawn 3 images as shown below.

Euclidean Distance

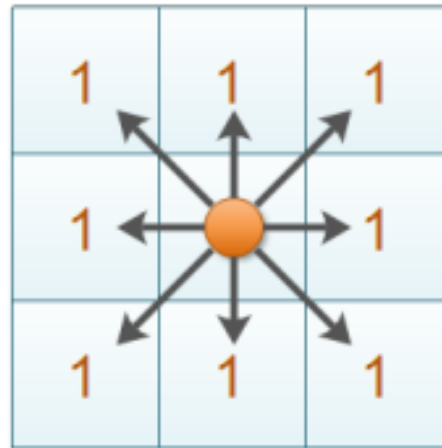


$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

The first one is Euclidean distance. The distance can be defined as a straight line between 2 points. The formula to calculate this has been shown in the image.

Other Distance Metrics (Euclidean, Chebyshev, and Manhattan Distances)

Chebyshev Distance

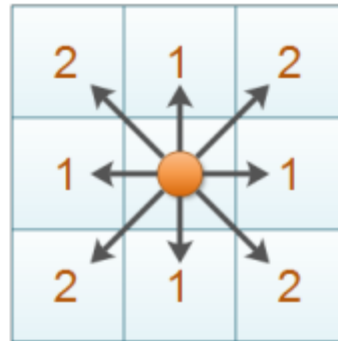


$$\max(|x_1 - x_2|, |y_1 - y_2|)$$

In Chebyshev distance, all 8 adjacent cells from the given point can be reached by one unit.

Other Distance Metrics (Euclidean, Chebyshev, and Manhattan)

Manhattan Distance



$$|x_1 - x_2| + |y_1 - y_2|$$

The last one is also known as L_1 distance. The distance between two points is the sum of the (absolute) differences of their coordinates. E.g. it only costs 1 unit for a straight move, but 2 if one wants to take a crossed move.

Non-Numeric Attributes

- Hamming Distance:
- It is defined as the number of attributes for which the two patterns have different values.

- Example:

P1: Habit = quiet, Eats = baked, Footwear = clogs

P2: Habit = gabby, Eats = baked, Footwear = clogs

Hamming Distance = 1 (quiet not equal to gabby for Habit attribute)

Numeric and Non-Numeric Attributes

- Example: Weather dataset: Outlook, Temperature and Pressure
- Hamming Distance for Non-Numeric Attribute (Outlook), Euclidean, Chebyshev, and Manhattan (Temperature, Pressure)
- Weighted sum of the these two distance makes the total distance measure

Strengths and Weaknesses

Advantages

- Robust to noisy training data (especially if we use inverse square of weighted distance as the “distance”)
- Effective if the training data is large

Disadvantages

- Need to determine value of parameter K (number of nearest neighbors)
- Distance based learning is not clear which type of distance to use and which attribute to use to produce the best results. Shall we use all attributes or certain attributes only?
- Computation cost is quite high because we need to compute distance of each query instance to all training samples. Some indexing (use K-D tree) may reduce this computational cost

Human Beings use Instance-based Learning

- Lot of research in this area (Dutt & Gonzalez, 2013; Gonzalez & Dutt, 2011; 2012)

Things to do

- Read Decision K Nearest Neighbor Tutorial by Teknomo (2011) (Material posted on Moodle)
- Read WEKA from Witten's book about KNNs
(<https://www.youtube.com/watch?v=zjYUYJ2b4r8>)
- Submit assignment



Questions or Comments?